

Question 15 – Garden Lights and Solar Cells

(a) Calculate the total resistance of the LED circuit

✓ 3 marks

Given:

- 10 LEDs connected in **parallel**
- Each LED + resistor has total resistance $R = 220\ \Omega$

For parallel circuits:

$$\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \dots (10 \text{ times}) = \frac{10}{220}$$

$$R_T = \frac{220}{10} = 22.0\ \Omega$$

✓ Total resistance is **22.0 Ω**

(b)(i) Name the effect

✓ 1 mark

✓ Answer: *The photoelectric effect*

(b)(ii) Explain how a potential difference is produced using band theory

✓ 3 marks

✓ Answer (band theory explanation):

- **Photons** incident on the **p-n junction** transfer energy to **electrons**
- Electrons gain enough energy to move from the **valence band** to the **conduction band**
- This creates **free electrons** and **holes**
- Movement of electrons and holes causes **charge separation**, producing a **potential difference**

🧠 Revision Tips

- In parallel: $R_T = \frac{R}{n}$
- A **p-n junction** forms the basis of solar cells, LEDs, and diodes
- Photons must have enough energy to promote electrons to the conduction band
- The **photoelectric effect** is the basis of **light-to-voltage conversion**

